Practical exercise 3: Fuzzy expert system to detect phishing in websites

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Abstract

Phishing websites pose a significant cybersecurity threat by imitating legitimate sites to trick users into disclosing sensitive information, such as credentials or financial data. The development of effective detection mechanisms is crucial to safeguarding online users. Artificial Intelligence (AI) techniquescan be employed to combat such cyber-attacks.

This project focuses on designing a simplified fuzzy expert system to classify webpages into four categories: safe, weakly suspicious, strongly suspicious, and phishing providing a numerical score between 0 and 100, representing the "phishing risk" level. Using a subset of five features from the 87 identified in existing literature, we define input linguistic variables with distinct scales and units.

Our approach uses rule-based reasoning through a Mamdani fuzzy inference system and validates the results with MATLAB's Fuzzy Toolkit. We define a manageable set of rules to decide whether a website is safe, suspicious, or phishing based on input variables. The system is tested with examples of websites, representing different risk scenarios, to ensure it behaves as expected. Each test is analyzed to show how the inputs lead to specific results. Finally, we propose a more advanced version of the system by adding more features.

1 Task 1: Definition of input & output linguistic variables.

Phishing websites employ various deceptive techniques to mimic legitimate websites and trick users into providing sensitive information. To detect such sites effectively, we selected five features, at least one from each of the categories from the paper proposed, which are URL-based features, Content-based features and external-based features. Below, we detail these features and their scales and units for each.

1.1 URL-Based Features

URL-based features analyze the structure and content of the web address to detect anomalies. These anomalies often indicate phishing attempts, as attackers manipulate URLs to deceive users. Among several URL-based features, we selected **Punycode**, as it is particularly relevant for phishing detection.

1. **URL Length:** Phishing websites often use excessively long URLs to disguise their true domain and mislead users. The goal is to obscure the main domain within a clutter of subdomains, directories, and query parameters, making it harder to identify the legitimacy of the website. Long URLs exploit users' tendency to skim over details, particularly when a URL appears overly technical or complex.

   For example, if someone registers `https://www.caixabank.com/login`, it is a concise and legitimate URL used by the bank for user login. However, a phishing attempt might use a URL like:

   `https://secure.caixabank.com.login-security-verify.info/verify/login`.

   At first sight, users might focus on the familiar term `caixabank.com` and ignore the actual domain, which in this case is `login-security-verify.info`.

   The deceptive prefix, `secure.caixabank.com`, is crafted to make the URL appear as if it belongs to the legitimate website, tricking users into believing they are on a secure page.

   The URL length is categorized into three linguistic variables defines as URL number of characters, defined using trapezoidal membership functions:

   (a) **Short URL**: [0, 0, 20, 30]

(b) **Medium URL**: [25, 50, 60, 80]

(c) **Long URL**: [70, 110, 180, 180]

1.2 Content-Based Features

Content-based features focus on analyzing the structure and elements of a webpage to detect patterns indicative of phishing. These features examine hyperlinks, forms, and other components that attackers often manipulate to mimic legitimate websites.

1. **Ratio of Null Hyperlinks:** Phishing websites often mimic target pages by replicating hyperlinks from legitimate sites but leaving them empty. This results in a higher ratio of null hyperlinks compared to legitimate sites.

   This feature is represented as a float value, calculated as the ratio of null hyperlinks to total hyperlinks. Higher values mean more probability of phishing while smaller ones indicate more legitimacy. The membership function for this feature is defined as a trapezoidal function with the following separations:

   (a) **Low Ratio**: [0.0, 0.0, 0.1, 0.3]

   (b) **Moderate Ratio**: [0.2, 0.3, 0.5, 0.7]

   (c) **High Ratio**: [0.6, 0.8, 1.0, 1.0]

2. **Login Forms:** Phishing attacks frequently involve login forms designed to steal user credentials. These forms often have external action links (redirecting to phishing servers) or empty actions (designed to capture input without a visible result). From these forms, attackers can collect sensitive information such as emails or phone numbers, which can be used to stage more targeted attacks, such as personalized phishing or social engineering attempts. Login forms with external action links or empty actions are considered phishing.

   This feature is represented as a binary value: 1 if the login form has external or empty action links (suspicious), and 0 otherwise.

1.3 External-Based Features

External-based features rely on metadata and information obtained from external services, such as domain registration details and search engine indexing. These features help detect phishing websites that are typically created for short-term use.

1. **Domain Age:** Legitimate websites are usually paid for and maintained over several years, reflecting their established and ongoing presence online. These websites often invest in long-term domain registrations to ensure continuity and build trust with users. In contrast, phishing websites are typically created for short-term use, as attackers aim to exploit users quickly before their activities are detected and blocked.

   This feature measures the age of the domain in months and is represented as an integer, where higher values represent more years and therefore more legitimacy. The membership function for this feature is defined with the following separations:

   (a) **New Site**: [0, 0, 6, 12]

   (b) **Recent Site**: [6, 12, 24, 36]

   (c) **Established Site**: [24, 48, 72, 96]

   (d) **Old Site**: [72, 120, 240, 240]

2. **Page Rank:** Page Rank is a numeric value that measures the importance or popularity of a webpage based on the number and quality of links to it. Developed by Google, it evaluates how well-connected a page is within the web. Legitimate websites generally have higher Page Rank values, reflecting their established and trusted presence online. Phishing websites, in contrast, often have low Page Rank scores due to their short lifespan and lack of meaningful connections to other trusted websites. This feature is represented as an integer value, with higher numbers indicating more trusted or influential pages. The membership function for this feature is defined with the following separations:

(a) **Poor Rank**: [0, 0, 1, 3]

(b) **Average Rank**: [2, 4, 6, 8]

(c) **High Rank**: [7, 9, 10, 10]

2 Task 2: Define the rules for the expert system.

The goal of this task is to define a set of conjunctive rules that will determine in which occasions a website is considered phishing or not according to the membership function. This membership functions for the phishing risk are defined as follows:

1. **Safe**: [0, 0, 15, 25]
2. **Weakly Suspicious**: [20, 30, 40, 50]
3. **Strongly Suspicious**: [45, 55, 65, 75]
4. **Phishing**: [70, 80, 100, 100]

The following set of conjunctive rules are designed to evaluate the phishing risk level of a website based on the selected features. Below are the rules defined for the expert system based on the selected features: URL Length, Domain Age, Page Rank, Ratio of Null Hyperlinks, and Login Forms.

The total number of possible rule combinations, based on these features, is $4 \times 3 \times 3 \times 3 \times 2 = 216$. This is calculated by multiplying the possible states of each feature. Domain Age has four possible memberships, Page Rank has three, URL Length has three, the Null Hyperlink Ratio has three and Login Forms have two possibilities. As a result, there are 216 unique combinations that represent all possible scenarios when evaluating the phishing risk of a website. Table 1 shows a summary of the rules used to cover all the combinations and the following sections explain the rules divided by the lenghts of the rules.

Age	**Pagerank**	**Login**	**URL Length**	**Hyperlinks**	**Output**	**Degree of Support**
-	-	Suspicious	Long	-	Phishing	0.95
-	-	Suspicious	-	High	Phishing	0.75
-	-	Suspicious	Not Short	High	Phishing	0.90
-	High	-	Short	Low	Safe	0.60
-	-	Normal	Short	-	Safe	0.70
Old	Not Poor	-	-	-	Weak Suspicious	0.40
New	Poor	-	Not Short	-	Strong Suspicious	0.60
Not Old	Average	Normal	-	-	Weak Suspicious	0.30
Old	Not Poor	Normal	-	-	Safe	0.90
-	-	Normal	Medium	Not High	Safe	0.60
Not New	Not Poor	Normal	-	-	Safe	0.75
-	-	Suspicious	-	Not Low	Phishing	0.40
-	-	Normal	Long	Not Low	Phishing	0.35
Old	High	-	-	-	Safe	0.70
-	-	Normal	Not Short	Low	Weak Suspicious	0.40
-	Not High	Suspicious	-	-	Phishing	0.60
-	High	Suspicious	Medium	Low	Weak Suspicious	0.80
-	-	Normal	Medium	High	Weak Suspicious	0.30

Table 1: Updated rules for phishing detection with corresponding degrees of support.

2.1 Two-Input Fuzzy Rules

The following rules describe phishing detection scenarios using combinations of two input features.

Rule 1: Explanation: Suspicious login forms combined with long URLs are a known tactic in phishing websites to obscure the actual domain and mislead users. These features suggest malicious intent, justifying a high degree of confidence in the classification.

1. **Inputs:**
    - **Login:** Suspicious
    - **URL Length:** Long
2. **Output:** Phishing
3. **Degree of Support:** 0.95

Rule 2: If the **Login** is *Suspicious* and the number of **Hyperlinks** is *High*, then the website is classified as **Phishing** with a degree of support of 0.75.
Explanation: Phishing websites often include a large number of hyperlinks to appear legitimate or to redirect users to malicious pages. When combined with suspicious login behavior, this pattern is strongly indicative of phishing activity.

1. **Inputs:**
    - **Login:** Suspicious
    - **Hyperlinks:** High
2. **Output:** Phishing
3. **Degree of Support:** 0.75

Rule 3: If the **Login** is *Normal* and the **URL Length** is *Short*, then the website is classified as **Safe** with a degree of support of 0.70.
Explanation: Normal login behavior and a short URL are typical indicators of a legitimate website. These features together provide moderate confidence in classifying the website as safe.

1. **Inputs:**
    - **Login:** Normal
    - **URL Length:** Short
2. **Output:** Safe
3. **Degree of Support:** 0.70

Rule 4: If the **Age** is *Old* and the **Pagerank** is *Not Poor*, then the website is classified as **Weak Suspicious** with a degree of support of 0.40.
Explanation: Older websites with a decent pagerank are less likely to be phishing, but some characteristics might still warrant caution, leading to a weak suspicion classification.

1. **Inputs:**
    - **Age:** Old
    - **Pagerank:** Not Poor
2. **Output:** Weak Suspicious
3. **Degree of Support:** 0.40

Rule 5: If the **Login** is *Suspicious* and the **Hyperlinks** are *Not Low*, then the website is classified as **Phishing** with a degree of support of 0.40.
Explanation: Suspicious logins combined with a high number of hyperlinks often point to phishing attempts, where malicious intent is higher, but the degree of suspicion remains moderate.

1. **Inputs:**
    - **Login:** Suspicious
    - **Hyperlinks:** Not Low
2. **Output:** Phishing

3. **Degree of Support:** 0.40
4. **EDG:** Moderate

Rule 6: If the **Age** is *Old*, the **Pagerank** is *High*, then the website is classified as **Safe** with a degree of support of 0.70.

Explanation: Websites that are old and have a high Pagerank are typically trustworthy, suggesting that the site has earned credibility over time.

1. **Inputs:**
   - **Age:** Old
   - **Pagerank:** High
2. **Output:** Safe
3. **Degree of Support:** 0.70
4. **EDG:** High

Rule 7: If the **Pagerank** is *Not High* and the **Login** is *Suspicious*, then the website is classified as **Phishing** with a degree of support of 0.60.

Explanation: Websites with a low Pagerank and suspicious login behavior are often used for phishing. This combination indicates a high likelihood of phishing attempts, although further factors are needed for full certainty.

1. **Inputs:**
   - **Pagerank:** Not High
   - **Login:** Suspicious
2. **Output:** Phishing
3. **Degree of Support:** 0.60
4. **EDG:** Moderate

2.2 Three-Input Fuzzy Rules

The following rules describe phishing detection scenarios using combinations of three input features.

Rule 8: If the **Login** is *Suspicious*, the **URL Length** is *Not Short*, and the number of **Hyperlinks** is *High*, then the website is classified as **Phishing** with a degree of support of 0.90.

Explanation: A suspicious login, a URL that is not short, and a high number of hyperlinks together point strongly to phishing. The URL length and hyperlink count amplify the suspicious nature of the login behavior, justifying a high confidence level.

1. **Inputs:**
   - **Login:** Suspicious
   - **URL Length:** Not Short
   - **Hyperlinks:** High
2. **Output:** Phishing
3. **Degree of Support:** 0.90

Rule 9: If the **Pagerank** is *High*, the **URL Length** is *Short*, and the number of **Hyperlinks** is *Low*, then the website is classified as **Safe** with a degree of support of 0.60.

Explanation: A high PageRank, short URL, and low number of hyperlinks are typical attributes of legitimate websites. This combination suggests the website is trustworthy, although with moderate confidence.

1. **Inputs:**

- **Pagerank:** High
- **URL Length:** Short
- **Hyperlinks:** Low

2. **Output:** Safe
3. **Degree of Support:** 0.60

Rule 10: If the **Age** is *New*, the **Pagerank** is *Poor*, and the **URL Length** is *Not Short*, then the website is classified as **Strong Suspicious** with a degree of support of 0.60.

Explanation: Newly created websites with poor pagerank and a URL that is not short often lack credibility and may indicate phishing attempts. These features combined lead to a strong suspicion classification.

1. **Inputs:**
   - **Age:** New
   - **Pagerank:** Poor
   - **URL Length:** Not Short
2. **Output:** Strong Suspicious
3. **Degree of Support:** 0.60

Rule 11: If the **Age** is *Not Old*, the **Pagerank** is *Average*, and the **Login** is *Normal*, then the website is classified as **Weak Suspicious** with a degree of support of 0.30.

Explanation: Websites with an average Pagerank and normal login status, combined with a not old age, raise weak suspicion due to the general characteristics of benign sites but lack distinctive features.

1. **Inputs:**
   - **Age:** Not Old
   - **Pagerank:** Average
   - **Login:** Normal
2. **Output:** Weak Suspicious
3. **Degree of Support:** 0.30
4. **EDG:** Low

Rule 12: If the **Age** is *Old*, the **Pagerank** is *Not Poor*, and the **Login** is *Normal*, then the website is classified as **Safe** with a degree of support of 0.90.

Explanation: Websites that are old and have a good Pagerank, along with normal login behavior, are generally trustworthy, suggesting a high degree of confidence in the safety classification.

1. **Inputs:**
   - **Age:** Old
   - **Pagerank:** Not Poor
   - **Login:** Normal
2. **Output:** Safe
3. **Degree of Support:** 0.90
4. **EDG:** High

Rule 13: If the **Login** is *Normal*, the **URL Length** is *Medium*, and the **Hyperlinks** are *Not High*, then the website is classified as **Safe** with a degree of support of 0.60.

Explanation: Normal logins and medium URL lengths with a reasonable number of hyperlinks suggest a benign website. The combination of these features leads to a moderate confidence in its safety.

1. **Inputs:**
    - **Login:** Normal
    - **URL Length:** Medium
    - **Hyperlinks:** Not High
2. **Output:** Safe
3. **Degree of Support:** 0.60
4. **EDG:** Moderate

Rule 14: If the **Age** is *Not New*, the **Pagerank** is *Not Poor*, and the **Login** is *Normal*, then the website is classified as **Safe** with a degree of support of 0.75.
Explanation: Websites that are not new, have a decent Pagerank, and a normal login behavior are generally safe, suggesting a relatively high degree of trustworthiness.

1. **Inputs:**
    - **Age:** Not New
    - **Pagerank:** Not Poor
    - **Login:** Normal
2. **Output:** Safe
3. **Degree of Support:** 0.75
4. **EDG:** High

Rule 15: If the **Login** is *Normal* and the **URL Length** is *Long* and the **Hyperlinks** are *Not Low*, then the website is classified as **Phishing** with a degree of support of 0.35.
Explanation: Websites with normal logins, long URLs, and many hyperlinks often attempt to obscure their real intentions, a common tactic in phishing schemes. These features increase suspicion, but with lower confidence compared to other phishing patterns.

1. **Inputs:**
    - **Login:** Normal
    - **URL Length:** Long
    - **Hyperlinks:** Not Low
2. **Output:** Phishing
3. **Degree of Support:** 0.35
4. **EDG:** Low

Rule 16: If the **Login** is *Normal* and the **URL Length** is *Not Short* and the **Hyperlinks** are *Low*, then the website is classified as **Weak Suspicious** with a degree of support of 0.40.
Explanation: A normal login with a not short URL and low hyperlinks suggests an unusual pattern that raises suspicion, but the overall confidence in classifying the site as suspicious is weak.

1. **Inputs:**
    - **Login:** Normal
    - **URL Length:** Not Short
    - **Hyperlinks:** Low
2. **Output:** Weak Suspicious
3. **Degree of Support:** 0.40
4. **EDG:** Low

Rule 17: If the **Login** is *Normal*, the **URL Length** is *Medium*, and the **Hyperlinks** are *High*, then the website is classified as **Weak Suspicious** with a degree of support of 0.30.

Explanation: Websites with a normal login and a medium URL length, along with many hyperlinks, do not immediately raise high suspicion. These characteristics suggest the site is potentially benign, but with weak suspicion due to the nature of its structure.

1. **Inputs:**
   - **Login:** Normal
   - **URL Length:** Medium
   - **Hyperlinks:** High
2. **Output:** Weak Suspicious
3. **Degree of Support:** 0.30
4. **EDG:** Low

2.3 Four-Input Fuzzy Rules

The following rules describe phishing detection scenarios using combinations of four input features.

Rule 18: If the **Pagerank** is *High*, the **Login** is *Suspicious*, the **URL Length** is *Medium*, and the **Hyperlinks** are *Low*, then the website is classified as **Weak Suspicious** with a degree of support of 0.80.

Explanation: A high Pagerank combined with suspicious login behavior, a medium URL length, and a low number of hyperlinks suggests a website that may appear trustworthy but still raises suspicion, especially with login irregularities.

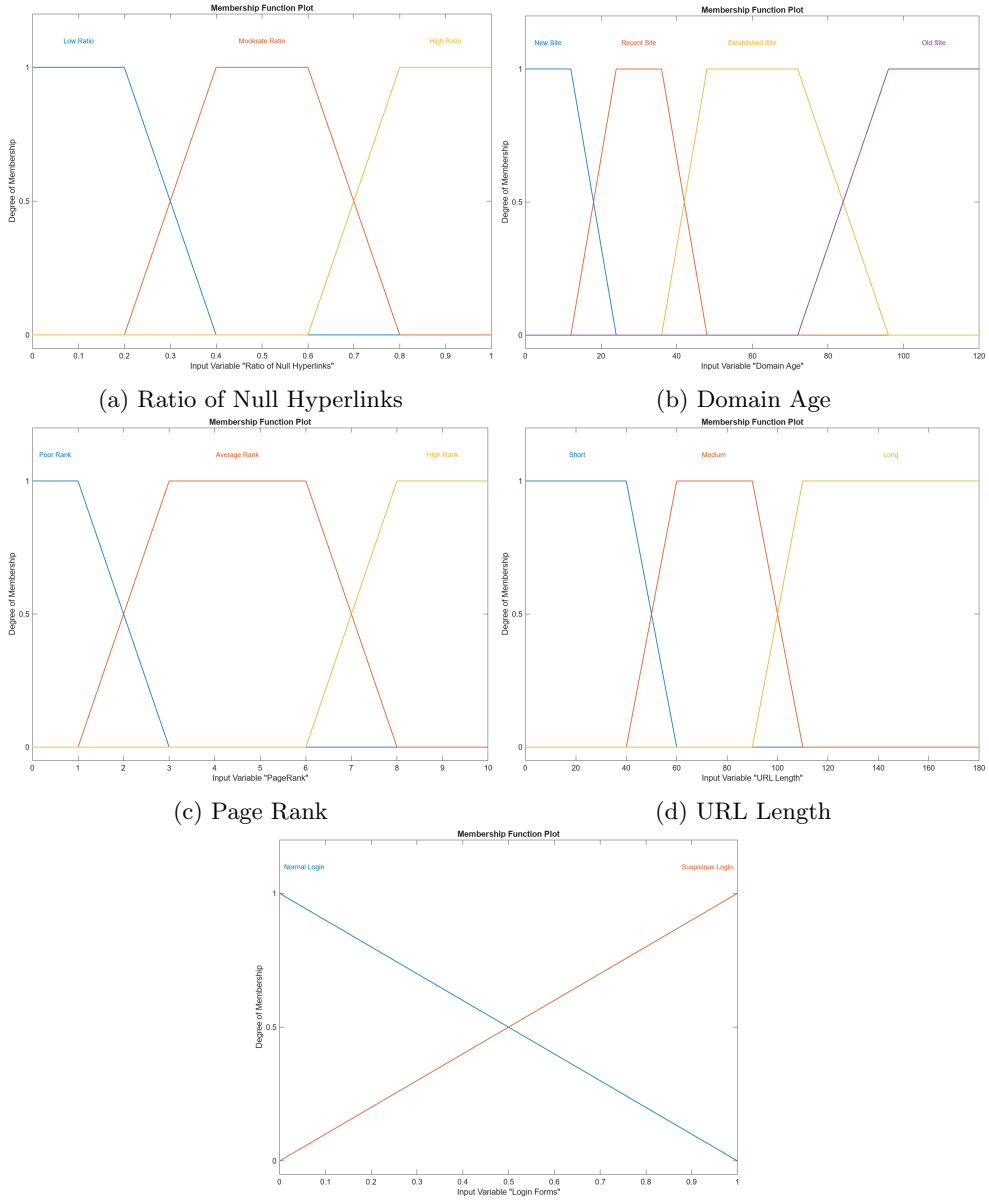
1. **Inputs:**
   - **Pagerank:** High
   - **Login:** Suspicious
   - **URL Length:** Medium
   - **Hyperlinks:** Low
2. **Output:** Weak Suspicious
3. **Degree of Support:** 0.80
4. **EDG:** Moderate

3 Task 3: Implementation of the system using Matlab.

The fuzzy expert system for web page classification was implemented using MATLAB and its Fuzzy Logic Toolbox. This environment provided a suitable platform for modeling, simulating, and evaluating the fuzzy inference system based on rules, which classifies web pages into the categories of Safe, Weakly Suspicious, Strongly Suspicious, and Phishing. The steps followed for the development of the system are described below:

The first step involved defining the problem and the variables involved. Five input variables were identified: Age (web page age), Pagerank (web page quality), Login (status of the login mechanism), URL Length (length of the URL), and Hyperlinks (number of hyperlinks). Additionally, an output variable named Phishing Risk was defined, which classifies websites into one of the four aforementioned categories. For further explanation see chapter 1.

Next, a new fuzzy inference system was created in MATLAB using the fuzzy command, which opens the FIS Editor. This editor allowed the configuration of an inference system based on the Mamdani model. Input and output variables were added to the system, and membership functions (MFs) were configured for each one. These functions were defined using triangular and trapezoidal shapes to ensure an accurate representation of the fuzzy sets. For instance, the variable Age included the sets New, Old, and Not Old, while the output variable Output included the categories Safe, Weakly Suspicious, Strongly Suspicious, and Phishing. The figure 1 shows each function.



(a) Ratio of Null Hyperlinks

(b) Domain Age

(c) Page Rank

(d) URL Length

(e) Login Forms

Figure 1: Membership function - Input Variables

The next step consisted of setting up the fuzzy rules that govern the system's behavior. These rules were defined based on the logical relationships between the input and output variables, for further explanation see chapter 2. These rules were implemented in MATLAB's rule editor or programmatically using the addRule function as shown in the figure 2.

Figure 3 illustrates a Mamdani fuzzy inference system developed to evaluate phishing risk based on five input variables: the ratio of null hyperlinks, domain age, PageRank, URL length, and the presence of login forms. Each input is modeled using Membership Functions (MFs) that classify values into categories such as low, medium, or high. These inputs are processed through a set of 18 fuzzy rules, which collectively determine the phishing risk output. The use of fuzzy logic allows the system to effectively handle uncertainty and provide a nuanced evaluation of risk.

To better understand how specific inputs influence phishing risk, Figures 4a, 4b, and 4c provide detailed visualizations.

Figure 4a examines the relationship between the ratio of null hyperlinks and domain age. The graph shows that phishing risk increases significantly as the proportion of null hyperlinks grows, with the system being particularly sensitive to this variable. The color gradient, ranging from blue (low risk) to yellow (high risk), helps visualize how these factors combine to impact the risk level.

	Rule	Weight	Name
1	If URL Length is Long and Login Forms is Suspicious Login then Phishing Risk is Phi…	0.95	rule1
2	If Ratio of Null Hyperlinks is High Ratio and Login Forms is Suspicious Login then Phi…	0.75	rule2
3	If Ratio of Null Hyperlinks is High Ratio and URL Length is not Short and Login Form…	0.9	rule3
4	If Ratio of Null Hyperlinks is Low Ratio and PageRank is High Rank and URL Length …	0.6	rule4
5	If URL Length is Short and Login Forms is Normal Login then Phishing Risk is Safe	0.7	rule5
6	If Domain Age is Old Site and PageRank is not Poor Rank then Phishing Risk is Wea…	0.4	rule6
7	If Domain Age is New Site and PageRank is Poor Rank and URL Length is not Short …	0.6	rule7
8	If Domain Age is not Old Site and PageRank is Average Rank and Login Forms is No…	0.3	rule8
9	If Domain Age is Old Site and PageRank is not Poor Rank and Login Forms is Norma…	0.9	rule9
10	If Ratio of Null Hyperlinks is not High Ratio and URL Length is Medium and Login For…	0.6	rule10
11	If Domain Age is not New Site and PageRank is not Poor Rank and Login Forms is N…	0.75	rule11
12	If Ratio of Null Hyperlinks is not Low Ratio and Login Forms is Suspicious Login then …	0.4	rule12
13	If Ratio of Null Hyperlinks is not Low Ratio and URL Length is Long and Login Forms …	0.35	rule13
14	If Domain Age is Old Site and PageRank is High Rank then Phishing Risk is Safe	0.7	rule14
15	If Ratio of Null Hyperlinks is Low Ratio and URL Length is not Short and Login Forms…	0.8	rule15
16	If PageRank is not High Rank and Login Forms is Suspicious Login then Phishing Ri…	0.6	rule16
17	If Ratio of Null Hyperlinks is Low Ratio and PageRank is High Rank and URL Length …	0.8	rule17
18	If Ratio of Null Hyperlinks is High Ratio and URL Length is Medium and Login Forms …	0.3	rule18

Figure 2: Rule Editor controller

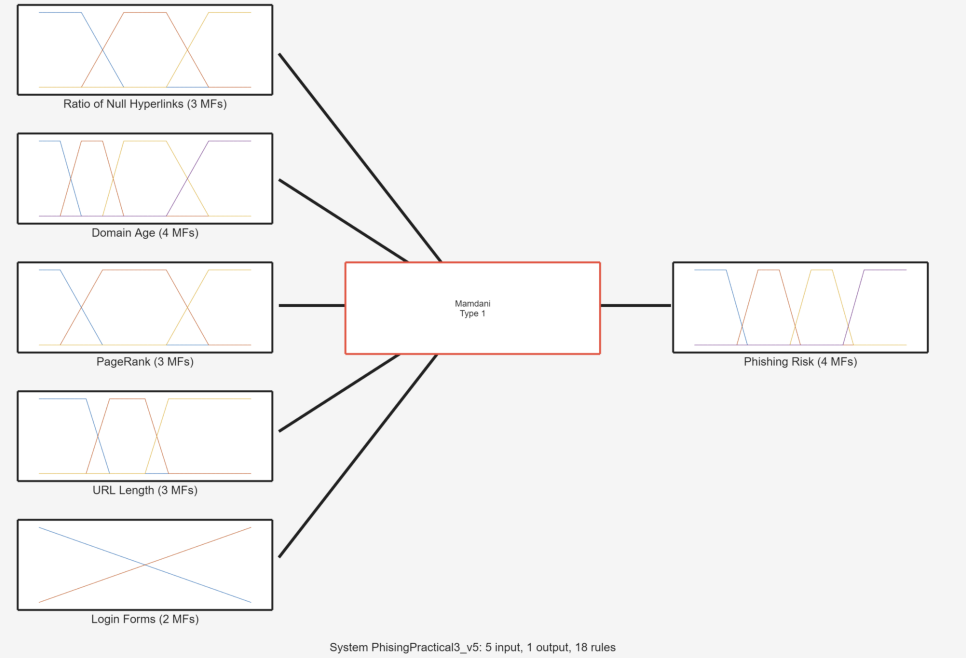
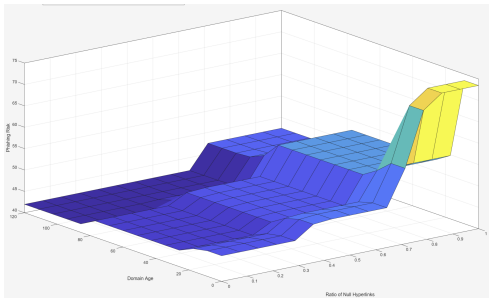


Figure 3: Input and output variables of the Fuzzy controller

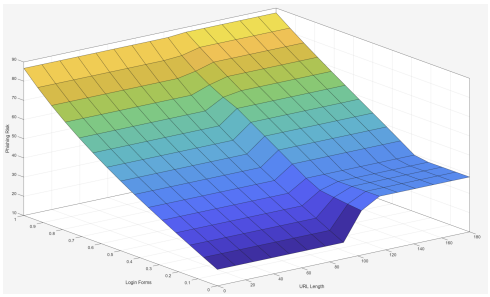
Building on this, Figure 4b explores the interaction between URL length and the presence of login forms. Both factors demonstrate a positive correlation with phishing risk, meaning that longer URLs and a higher number of login forms contribute to an increased likelihood of phishing. This graph also uses a color spectrum to convey the risk levels, making it easier to interpret how these variables influence the system's output.

To provide a more comprehensive view, Figure 4c presents a 3D visualization of the interplay between the ratio of null hyperlinks and the presence of login forms. In this plot, the X-axis represents the ratio of null hyperlinks, the Y-axis depicts the number of login forms, and the Z-axis indicates the calculated phishing risk. The surface plot highlights that a combination of a high proportion of null hyperlinks and login forms significantly raises the phishing risk.

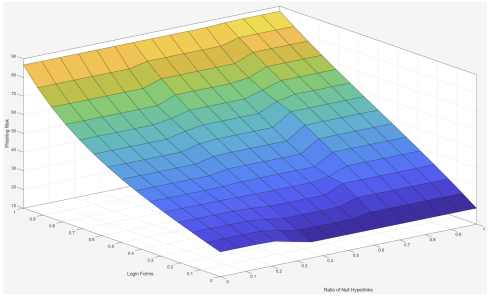
These visualizations collectively offer valuable insights into the relationships between the input variables and their effect on the system's output. They enable a clear understanding of trends and patterns, which can be used to refine the fuzzy rules, thereby improving the system's accuracy in identifying phishing risk.



(a) Ratio of Null Hyperlinks



(b) Domain Age



(c) Page Rank

Figure 4: Membership function - Input Variables

Once the system was configured, it was validated. The fuzzy model was tested with different combinations of input data using the evalfis, which calculates the fuzzy output for a specific set of inputs. This allowed verification that the obtained outputs matched the expectations based on the defined rules.

Finally, the system was applied to a test dataset to evaluate its performance under real-world conditions. The results will be presented in the next chapter.

4 Task 4: Test the system with four different websites.

4.1 Test 1

This section presents the analysis of the first case evaluated using the fuzzy expert system. The input values considered for this problem are the following:

Input

- URL Length: 30 (Short URL)
- Ratio of Null Hyperlink: 0.1 (Low Ratio)
- Login Forms: 0 (No Login Form)
- Domain Age: 120 months (Old Site)

- Page Rank: 8 (High Rank)

The result obtained after evaluating the 18 rules defined in the system was 18.9. This value represents the fuzzy phishing risk score calculated based on the input values and the inference of the system's rules.

The obtained value (18.9) belongs to the **Safe** category as it falls within the range [15, 25]. This indicates that the system considers this webpage to be safe, with a very low risk level. Furthermore, there is no significant influence from the categories *Weakly Suspicious*, *Strongly Suspicious*, or *Phishing*, since the value is far from their respective ranges.

4.2 Test 2

This section presents the analysis of the second case evaluated using the fuzzy expert system. The input values considered for this problem are the following:

Input

- URL Length: 55 (Medium URL)
- Ratio of Null Hyperlink: 0.4 (Moderate Ratio)
- Login Forms: 1 (Login Form)
- Domain Age: 90 months (Old Site)
- Page Rank: 5 (Average Rank)

The result obtained after evaluating the 18 rules defined in the system was 69.7. This value represents the fuzzy phishing risk score calculated based on the input values and the inference of the system's rules.

The obtained value (69.7) is very close to the boundary of the **Phishing** category, as it falls slightly below the range [70, 80]. However, the proximity suggests that this webpage has a high likelihood of being classified as strongly suspicious or phishing. The result highlights a considerable risk level, influenced by factors such as the moderate ratio of null hyperlinks and the presence of a login form.

The analysis of this second case indicates that the webpage under evaluation presents a high level of risk, leaning towards the **Phishing** category. This result underscores the significance of inputs such as login forms and hyperlink ratios in determining phishing risks. The suggested visualizations further validate the system's reasoning and ensure the reliability of the defined fuzzy rules.

4.3 Test 3

This section presents the analysis of the third case evaluated using the fuzzy expert system. The input values considered for this problem are the following:

Input

- URL Length: 80 (Long URL)
- Ratio of Null Hyperlink: 0.7 (High Ratio)
- Login Forms: 1 (Login Form)
- Domain Age: 1 months (New site)
- Page Rank: 2 (Poor Rank)

The result obtained after evaluating the 18 rules defined in the system was 78. This value represents the fuzzy phishing risk score calculated based on the input values and the inference of the system's rules.

The obtained value (78) belongs to the **Phishing** category as it falls within the range [70, 80, 100, 100]. This indicates that the system identifies this webpage as highly suspicious and likely to be a phishing site. The high value suggests significant influence from the *Long URL*, *High Ratio of Null Hyperlinks*, and *New Domain Age*, which are strong indicators of phishing attempts.

4.4 Test 4

This section presents the analysis of the fourth case evaluated using the fuzzy expert system. The input values considered for this problem are the following:

Input

- URL Length: 150 (Very Long URL)
- Ratio of Null Hyperlink: 1 (Very High Ratio)
- Login Forms: 1 (Login Form)
- Domain Age: 0.2 months (New site)
- Page Rank: 0 (Poor Rank)

The result obtained after evaluating the 18 rules defined in the system was 77.4. This value represents the fuzzy phishing risk score calculated based on the input values and the inference of the system's rules.

The obtained value (77.4) belongs to the **Phishing** category as it falls within the range [70, 80, 100, 100]. This indicates that the system classifies this webpage as highly suspicious and very likely to be a phishing site. The high score is mainly driven by the combination of a *Very Long URL*, a *Very High Ratio of Null Hyperlinks*, a *New Domain Age*, and a *Poor Page Rank*. These factors are strong indicators commonly associated with phishing attempts.

5 Task 5: Design of a complete fuzzy expert system.

To design a more complete fuzzy expert system, we are going to include more features and some extra rules that may prove useful in identifying phishy websites. We will add more features from each of the categories we have used for task 1, also coming from the paper proposed.

5.1 URL-Based Features

1. **Special Characters**: the number of special characters that a URL contains can be an indicator of phishing, since they can be used to deceive users. Special characters are points, commas, question marks, tides, ampersat and so on.

   This feature is an integer with the following trapezoidal memberships:

   - Few Special Characters: [0, 0, 2, 5]
   - Some Special Characters: [3, 5, 6, 8]
   - Quite some Special Characters: [6, 8, 10, 12]
   - a Lot of Special Characters: [10, 12, 15, 15]

2. **HTTPS Token**: Phishing websites tend to lack any security facilities, therefore the HTTPS token may be missing, whereas legitimate do have one. This feature is a binary feature, where one means the URL has a HTTPs token and 0 means it does not.

5.2 Content-Based Features

1. **Pop-Up Window**: Phishing websites often tend to use pop-up windows to introduce information, while legitimate websites usually only use them as warnings. Therefore, they are a good indicator to know if a website is legitimate. This feature is a binary feature, where one means the website has a pop-up window with text fields and 0 means it does not.

5.3 External-Based Features

1. **Web traffic**: The number of people who visit the web tends to be higher in legitimate websites, therefore, the number of people that visits the site in one day can be a grate indicator to detect phishing.

   This feature is an integer with the following trapezoidal memberships:

- Few Visitors Characters: [0, 0, 10, 20]
- Some Visitors Characters: [15, 25, 35, 45]
- a Lot of Visitors Characters: [40, 55, 70, 70]

2. **DNS record**: Domain Name Server (DNS) is mandatory to retrieve the IP address of URLs for access. This means that websites must have the URL registered, therefore legitimate websites always have one, whereas phishing websites may fail to have one. This feature is a binary feature, where one means the website is registered and 0 means it is not.

3. **WHOIS Registered Domain**: if a page is not registered in WHOIS, it will likely be a phishing website, seeing as most legitimate websites usually appear in its records. This feature is a binary feature, where one means the website is registered and 0 means it is not.

Adding all these extra features to the model (5) would lead to a more complete fuzzy expert system. Whatsoever, new rules should also be added to the system for it to be more complete. Some rules that would make sense to include would be:

- Page rank true and domain age high implies a legitimate website
- Domain Age low and WHOIS domain missing would be an indicator of phishing.
- URL length short and DNS record true is pretty much a legitimate website
- Web with few visitors and a Poor Rank would mean a phishing website.

Adding these new features and new rules would give way to a more accurate fuzzy expert system to identify phishing.

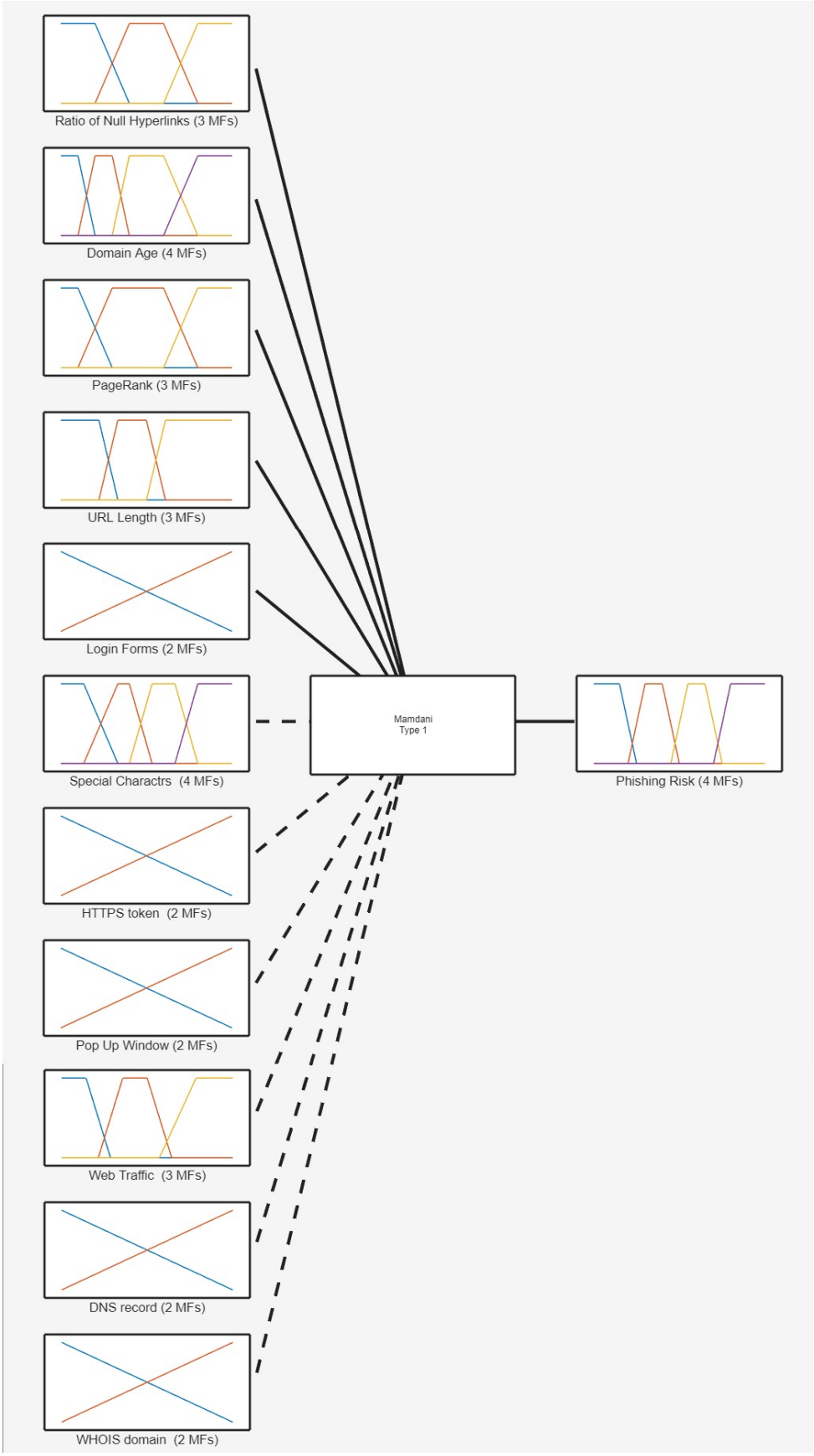


Figure 5: Input and output variables in Complete Fuzzy System

6 Conclusions

The evaluation of the four test cases using the fuzzy expert system provided valuable insights into its capability to classify webpages based on phishing risk. The following conclusions can be drawn from the results:

- The fuzzy expert system effectively identified different risk levels across the tested cases, demonstrating its sensitivity to input variations such as URL length, hyperlink ratio, login form presence, domain age, and page rank. The calculated risk scores closely aligned with the expected outcomes based on the defined fuzzy rules, confirming the system's robustness.
- Certain inputs, such as the ratio of null hyperlinks, domain age, and page rank, significantly influenced the classification. For example, cases with a higher ratio of null hyperlinks and newer domain ages consistently produced higher risk scores. These findings highlight the importance of these variables in identifying phishing sites.
- The system reliably categorized webpages into predefined fuzzy categories: Safe, Weakly Suspicious, Strongly Suspicious, and Phishing. This demonstrates its practical applicability in real-world scenarios, as it successfully flagged pages with high-risk attributes, such as very long URLs, very high null hyperlink ratios, and poor page ranks, as phishing threats.

About each test:

- Test 1: The webpage was correctly classified as Safe, supported by strong factors like a short URL, low null hyperlink ratio, and high page rank.
- Test 2: Although the risk score fell slightly below the Phishing range, the system flagged it as strongly suspicious due to moderate hyperlink ratios and the presence of a login form.
- Test 3: The system categorized the webpage as Phishing due to high-risk factors, such as a long URL, high null hyperlink ratio, and a new domain age.
- Test 4: The system identified this case as Phishing, heavily influenced by extreme risk indicators like a very long URL, very high null hyperlink ratio, a new domain age, and a poor page rank.

In summary, the fuzzy expert system demonstrated reliable performance in classifying webpages by phishing risk. Its interpretability and adaptability make it a promising tool for mitigating cybersecurity threats. Further enhancements could extend its effectiveness in dynamic, real-world environments.